



South East Asian Institute of Technology, Inc
College of Information and Communication Technology
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A Research Project on

SEAIT Anti-Cheating Quiz: AI-Driven Adaptive Quiz System with Anti-Cheating Detection for Improving College Learning and Assessment at South East Asian Institute of Technology Inc.

As partial requirement for the Subject

IT ELEC 4

Advanced Human Computer Interaction 2

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1.0 INTRODUCTION

1.1 Background and Context

Human–Computer Interaction (HCI) is a field of research that studies how people interact with computers and digital technologies. It combines ideas from design and human behavior to develop systems that are usable, efficient, and user-friendly. Advancements in HCI have made technology more accessible, especially in educational environments where timely and effective feedback plays an important role in the learning process (Sarma & Bhuyan, 2021). Over time, HCI has evolved from early command-line interfaces to modern graphical user interfaces (GUIs) and touch-based interactions, greatly influencing how users experience mobile devices, computers, and other emerging technologies. This research focuses on the development of the SEAIT Anti-Cheating Quiz system, which integrates adaptive questioning and behavior monitoring to support academic integrity and student engagement. The system aims to improve assessment fairness, provide meaningful feedback, and create an interactive user experience guided by HCI principles.

1.2 Research Problem

This study aims to determine the effectiveness of the SEAIT Anti-Cheating Quiz system in improving assessment and student engagement. Many existing online quiz platforms lack adaptive questioning and effective behavior monitoring tools, which may result in academic dishonesty and reduced student interaction. Furthermore, some monitoring systems negatively affect usability and user satisfaction. This research seeks to measure the impact of adaptive questioning and suspicious behavior pattern detection on student engagement, usability perception, and assessment reliability through quantitative data analysis.

1.3 Research Questions and Objectives

Research Questions:

- How does the adaptive questioning feature of the SEAIT Anti-Cheating Quiz system influence student engagement in online assessments?
- How does suspicious behavior detection affect the overall usability and satisfaction of students using the system?
- How effective is the SEAIT Anti-Cheating Quiz system in maintaining fairness in online assessments compared to traditional quiz platforms?

Research Objectives:

- To determine the effect of adaptive questioning on student engagement in online assessments.
- To determine the impact of suspicious behavior detection on usability and student satisfaction.
- To evaluate the effectiveness of the system in maintaining fairness in online assessment.

1.4 Justification and Significance

This study addresses Human–Computer Interaction (HCI) challenges in online assessment systems, particularly in ensuring usability, student engagement, and fairness in assessments. The SEAIT Anti-Cheating Quiz system integrates adaptive questioning and suspicious behavior detection to create a more secure and interactive assessment environment for college students. As online quizzes become increasingly common in higher education, concerns regarding cheating, system usability, and student participation continue to arise. This research aims to improve digital assessment experiences by evaluating how adaptive and monitoring features can enhance engagement while maintaining fairness and reliability. The findings of this study will contribute to HCI knowledge by identifying user-centered design principles and usability in educational technology. The results may benefit students through improved interaction and engagement, instructors through better monitoring and performance insights, and institutions through data-driven recommendations for secure and user-friendly assessment platforms.

2.0 LITERATURE REVIEW

2.1 Overview of HCI Theories and Models

Recent advances in Human–Computer Interaction (HCI) emphasize usability as a core principle in educational technology design. A state-of-the-art systematic review conducted in 2022 analyzed usability research in educational and learning technologies over the past two decades. The review revealed that most usability studies focused primarily on technological aspects, such as interface design and functionality, while pedagogical and socio-cultural dimensions were often overlooked. Additionally, conceptual and methodological limitations were identified, suggesting that many usability evaluations may produce unreliable or incomplete results. These findings demonstrate the evolving nature of HCI research, shifting from purely technical usability concerns toward more comprehensive, user-centered and context-aware approaches in educational systems.

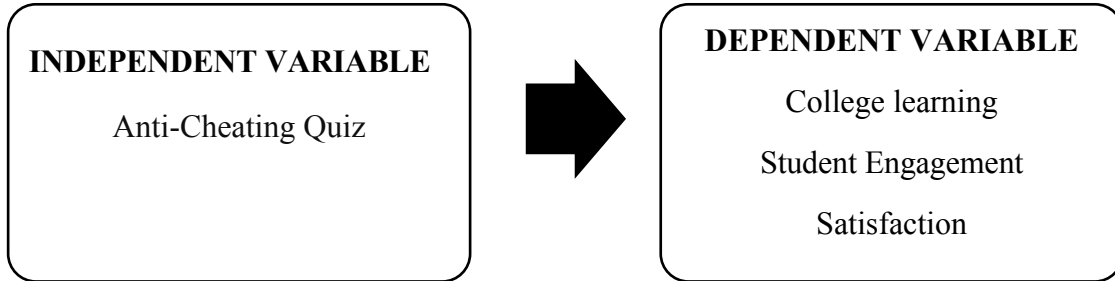
2.2 Review recent studies, papers, and advancements in HCI

Recent studies have explored the use of artificial intelligence and computer vision to detect cheating in online exams. Essahraoui et al. (2022) proposed an intelligent anti-cheating model that analyzes students' postures in real-time using CNN-Facial-landmark and MediaPipe models, combined with an LSTM classifier to predict potential cheating behaviors. The model achieved 94% accuracy on the training set and 75% on the test set, demonstrating the feasibility of automated behavior monitoring in educational assessments. While effective, the study highlights potential challenges such as false positives, usability concerns, and the need to balance monitoring with student privacy. This research represents a recent advancement in HCI for educational systems, emphasizing the integration of AI-driven solutions to enhance assessment integrity.

2.3 Analyze existing solutions related to the research problem

Many AI-based anti-cheating systems have been developed to improve integrity in online and lab-based assessments. These systems use behavior analysis, facial recognition, and adaptive questioning to monitor students and prevent dishonest behavior. However, some limitations still exist. For example, certain cheating methods are not detectable, such as students using multiple devices, consulting off-camera resources, or bringing physical notes during lab quizzes (Gloobia, 2023).

Conceptual Framework



3.0 METHODOLOGY

3.1 Research Design

This study will use a descriptive quantitative research design to determine participants' experiences using the SEAIT Anti-Cheating Quiz system. The independent variable will be the quiz system, while the dependent variables include student engagement, assessment integrity, and satisfaction. This design is appropriate for providing a clear summary of students' experiences and evaluating the effectiveness of the anti-cheating features in supporting fair and engaging online assessments.

3.2 Participants

The participants of this study will be students at the South East Asian Institute of Technology, Inc., taking courses that require online or lab-based quizzes. A total of 100 students will be invited to participate. The study will measure student engagement, assessment integrity, and overall satisfaction, allowing evaluation of how the SEAIT Anti-Cheating Quiz system affects student experiences during assessments.

3.3 Data Collection

Quantitative data will be gathered through structured questionnaires designed to assess student engagement, overall of assessment integrity, and satisfaction with the SEAIT Anti-Cheating Quiz system. Random sampling will be applied to ensure that data represent the target population. The data collected will allow for statistical analysis, enabling identification of trends and patterns in student responses and their coconnection with system features such as adaptive questioning and anti-cheating.

3.4 Data Analysis

Data gathered from the structured questionnaires will be analyzed using descriptive statistics, including means, standard deviations. Analysis will focus on student engagement, perception of assessment integrity, and overall satisfaction with the system, providing insight into the effectiveness and usability of the SEAIT Anti-Cheating Quiz system.

3.5 Ethical Considerations Indention

The study will prioritize participant privacy and data protection. Will be obtained from all participants, who may withdraw at any time. Any reported results will not disclose participant identities. Researchers will follow ethical guidelines for HCI and educational technology research to ensure participants do not experience harm, discomfort, or breach of privacy during the study.

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